

Braess Paradox in Layer-2 Blockchain Payment Networks

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Abstract—Payment channel networks (PCNs) are a leading method to deal with the scalability limitations of blockchain networks. PCNs allow users to execute transactions without committing them to the blockchain by relying on predefined payment channels. Transactions between pairs of users without a connecting channel are also supported through a path of multiple channels. Serving such transactions involves fees paid to intermediate users. In this paper, we uncover the potential existence of the Braess paradox in payment networks: Sometimes, establishing a new payment channel can increase the fees paid for serving some fixed transactions. We study conditions for the paradox to appear under two different models of fees: Liquidity-based fees as a function of the transaction value jointly with the channel liquidity values and proportional fees which are based only on the transaction value. We also provide indications for the appearance of the paradox based on real data from Lightning, a popular payment channel network of Bitcoin. Last, we discuss methods to mitigate the paradox upon establishing a new payment channel.

Index Terms—Blockchain, Payment Channel Networks, Braess Paradox.

I. INTRODUCTION

A. Payment Channel Networks

Blockchain networks like Bitcoin and Ethereum suffer from scalability issues, having limited transaction rates of up to tens of transactions per second, while, for instance, credit cards like Visa and Mastercard can handle thousands of transactions per second [1]–[3]. This motivated the rise of payment channel networks (PCNs) like Bitcoin’s Lightning [4] as well as the TeeChain network [5] or the Perun system [6], all allowing much higher transaction rates. PCNs rely on payment channels between pairs of nodes without recording each transaction over the blockchain.

To establish a payment channel, two users create an on-chain transaction in which they lock amounts of money that determine the initial channel liquidity values. The users can then perform mutual transactions quickly, without updating the blockchain regarding every transaction. Such transfers move the balance from the channel liquidity of one user towards that of the other. Transfers are available in both directions as long as both liquidity values remain nonnegative.

Creating a payment channel might be an expensive action as it requires an on-chain transaction as well as locking money in the channel. A user can execute a transaction with other users even if they are not connected by a direct channel but through a path of multiple channels involving other intermediate users. In such cases, *fees have to be paid* by the user establishing

the transaction. Eventually, when users want to close their payment channel, they report to the blockchain the updated balance of each user after serving the series of transactions.

B. Braess Paradox

Braess paradox [7] is a counterintuitive network phenomenon where adding a link to a network potentially increases the travel time in the network, although such an additional link adds to flexibility in routing. Braess paradox was shown to appear in various areas, including transportation [7], communication networks [8], power grid energy [9] and biological systems [10]. Interestingly, it has been shown that in the context of blockchain constant function market makers (CFMMs), Braess paradox cannot appear when trading [11]. Although typically Braess paradox appears in graphs of a particular structure, Valiant and Roughgarden [12] showed that large random graphs often have instances of subgraphs with such structural properties. In this work, we reveal and analyze the potential existence of Braess paradox in blockchain payment channel networks. We refer to the *Braess paradox in payment channel networks* as the occurrence of a fee increase following the establishment of a new payment channel. The paradox is expressed as increased fees upon serving transactions following the establishment of a new payment channel.

Fig. 1 illustrates the classic case of Braess Paradox in a transportation network: Each link describes a directed road, and link weights refer to the time it takes to cross a link: Either as a fixed time or as a function of the number x of drivers crossing the link. Assume there are 40 drivers who wish to get from *start* to *end*, and initially, road e_{23} (between nodes s_2, s_3) does not exist. At an equilibrium state, 20 of the drivers choose path $p_{up} = start \rightarrow s_2 \rightarrow end$ and the other 20 take $p_{down} = start \rightarrow s_3 \rightarrow end$. The time it takes each driver

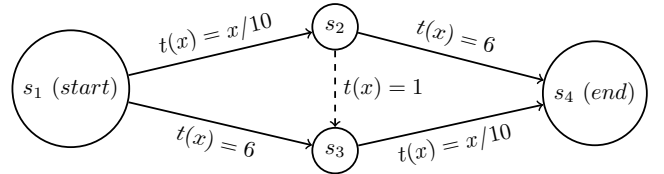


Fig. 1: Illustration of Braess Paradox in a transportation or communication network. Traffic is sent from the left node s_1 (*start*) to the right node s_4 (*end*). $t(x)$ describes the time (e.g., in minutes) it takes to cross a link where x refers to the amount of traffic over the link. The impact of the addition of the dashed link $s_2 \rightarrow s_3$ is examined.

TABLE I: The Existence of the Braess Paradox - Contributions overview

Number of Rounds	Fee Model		
	Fixed	Proportional to transaction size	Liquidity-based
Single round	does not exist Lemma 1, Property 1	does not exist Property 1	can exist Theorem 1
Multiple rounds	can exist Corollary 1	can exist Theorem 2	can exist Lemma 3, Lemma 4

to get from *start* to *end* is $t = \frac{20}{10} + 6 = 8$ minutes, and it is not worth it for any of the drivers to change their strategy through selecting an alternative path. Next, the addition of e_{23} is examined. Now, drivers who chose p_{up} can improve their strategy by taking $p_{mid} = start \rightarrow s_2 \rightarrow s_3 \rightarrow end$ as there are at most 40 drivers hence the time of $s_2 \rightarrow s_3 \rightarrow end$ is necessarily shorter than that of $s_2 \rightarrow end$. Similarly, drivers who use the path p_{down} can improve their strategy by also choosing p_{mid} , as the time of $start \rightarrow s_2 \rightarrow s_3$ must be shorter than that of $start \rightarrow s_3$. Accordingly, all drivers now use p_{mid} so it takes each driver $t = \frac{40}{10} + 1 + \frac{40}{10} = 9$ minutes. Note that this is an equilibrium state as no driver can reduce their travel time by changing their path. Overall, adding link e_{23} increases the traveling time from 8 to 9 minutes.

To the best of our knowledge, this paper is the first to present and discuss Braess paradox in payment channel networks. Our model is based on real-world liquidity-dependent fee functions and channel designs used in contemporary payment channel networks. The paradox we present shares similarities with classical models that analyze the Braess paradox in various domains. However, we also identify several features that are unique to payment channel networks:

- (i) The main performance metric refers to the cost of fees rather than latency.
- (ii) While in regular communication networks, paradox appearance depends on the current network traffic, in payment channel networks the paradox relies also on historical traffic (previous transactions).
- (iii) Unlike regular communication networks, the paradox's existence in payment channel networks depends not only on the number of transactions in the network but also on their content (the values of the transactions).

We study the appearance of the paradox based on the three forms of fees: Fixed, proportional to the transaction size or based on the updated liquidity of the channel. We also distinguish between a single payment and multiple rounds. In addition to the theoretical analysis, we show the existence of the multi-round paradox on Bitcoin's Lightning network. Finally, we discuss how the paradox can be mitigated.

In the paper, we use several assumptions backed by real-life scenarios and data:

- We assume several channels have high liquidity while others suffer from low liquidity, similar to real-life data of Lightning (Fig. 8).
- We assume greedy algorithms are used by users to find

the cheapest existing route, similar to existing routing algorithms (Section II-D).

- We assume the graph follows the classic Braess-paradox-graph presented in Fig. 1. This graph is used to illustrate the well-known phenomenon of Braess paradox, which is known on this graph. Moreover, Section IV-D shows a reduction from general complex real-life network graphs to this simple four-node graph. This allows for an easy-to-follow analysis with simple and realistic conditions and rules of thumb that can be applied to large real-life networks.

Table I summarizes the findings regarding the potential existence of the paradox under the various scenarios.

C. Related Work

Routing in Payment Channel Networks. Previous works have studied routing schemes for payment networks. Flash [13] categorizes transactions based on their value as mice and elephants. To balance between high throughput and probing overhead, it simply routes the small transactions based on random route selection. Spider [14] combines transactions and uses a multi-path congestion control transport protocol to ensure balanced utilization of channels and fairness across pairs of users. Flare [15] is a routing scheme for Bitcoin's Lightning network where nodes share their knowledge of the network in the form of a routing table used to find paths. Survivable payment channels [16] proposes a method for capacity distribution among network payment channels to increase the time until any channel becomes unidirectional. SpeedyMurmurs [17], Voute [18] and HushRelay [19] are privacy-preserving routing schemes that consider the initial and dynamic balance of channels. Similarly, Simões et al. studied improving privacy through mixing and indirect payments [20].

Topology Design for Payment Channel Networks. Demand-aware designs of payment channel topologies were studied. An existing recent approach [21] studied two problems: Constructing a topology of a limited number of channels that can effectively serve a demand for transactions, or mapping users to a given topology of a known structure such as a star, ring, or expanders. Some previous works also used game theory concepts for network design. For instance, Avarikioti et al. [22] analyzed Nash Equilibrium from the view of the user who intends to transmit money and can either use an existing path or open a new direct payment channel with its involved

costs. Ersoy et al. [23] studied fees maximizing the revenue to channel owners while keeping them attractive to use.

Properties of Payment Channel Networks. We study the question: *Can adding a new payment channel to a payment channel network increase the fees needed to be paid?* An equivalent question would be: Does the existence of a channel cause a fee increase? To the best of our knowledge, we are the first to study this important question.

Previous papers tried to answer a similar question: Can adding a new channel decrease transaction success rate and harm the throughput of the system? Since on-chain fees are usually much higher than off-chain fees, decreasing transaction success rate might not only harm transaction throughput but might also cause a fee increase. [24] discusses this question and shows that under a natural model of repeated transactions and the assumption that the transaction rate matrix is symmetric, the probability that two users can successfully transmit a transaction cannot decrease when a new channel is added to the network. Under this model, [25] continues in this direction and presents equations by which users can compute the probability of transaction success in different network states. In fact, as we show in this paper, even without decreasing the success rate, adding a new channel might cause a fee increase.

Paper outline: First, in Section II we describe the model of the paper, present basic properties of the Braess paradox and provide an overview of common fee functions. In Section III we refer to liquidity-based fee functions for which we present the Braess paradox in a single round and analyze the conditions for its appearance. Next, in Section IV we also refer to proportional fee functions for which we analyze the multi-round paradox. We show examples of such cases and discuss the conditions for the paradox appearance under different models and graph structures. In Section V, we present statistics supporting our assumptions and provide early indications for the multi-round paradox appearance in the Lightning network, using several routing functions. Finally, in Section VI, we propose possible solutions for mitigating the paradox.

II. OVERVIEW

A. Model and Notations

We model a payment channel network as a directed graph where nodes represent users, and a link e_{ij} refers to a directed payment channel between nodes s_i and s_j . Table II summarizes the main notations of the paper.

Two users u_1, u_2 establish a channel by locking values of c_1, c_2 such that $c_1, c_2 \geq 0$ and potentially where $c_1 \neq c_2$ with a total locked amount of $c_1 + c_2$. We refer to (c_1, c_2) as the (initial) channel liquidity values. When u_1 transfers an amount Δ to u_2 , the channel liquidity values change to $(c_1 - \Delta, c_2 + \Delta)$. Although opening a channel between users may allow transfers in both directions, we view a channel as two individual parallel links, directed in opposite directions. In some payment channel networks (such as Lightning and Raiden) each channel has its own fee policy, and channels may be disabled in one direction (thus, in practice, such channels will be represented by only one directed link). Transactions

TABLE II: Summary of main notations

Symbol	Meaning
x	value of a transaction
Φ	transaction fee function
F	channel's liquidity
γ, δ	fee function proportional and base rates parameters
α	normalization factor
k	number of transaction rounds
s_i	a PCN node
e_{ij}	a directed payment channel between nodes s_i and s_j

can be served as long as the liquidity values in a channel remain nonnegative.

Serving a transaction involves a cost, given as the sum of fees along the channels through which the transaction is routed. The nodes along such a route are called intermediate nodes. Upon using a channel between two nodes (in a particular direction), the source node charges a fee. In Lightning, intermediate nodes charge a fee of the form $\Theta(x) = \gamma \cdot x + \delta$ where $\gamma \cdot x$ is a proportional fee dependent on the transaction value x and δ is a base fee [26]. In Raiden, the fee function additionally includes an imbalance fee that increases when the transaction exhausts channel liquidity. This aims to keep the channel balanced [27]. Additionally, as channel liquidity is limited, fee functions based on the channel's current liquidity [28]–[30] are intuitive. The liquidity of a channel and, accordingly, its fee change over time, based on the transactions it serves. Section II-C provides more information regarding liquidity-based fees.

As a channel is limited by its liquidity, it may become unusable over time, depending on the liquidity balance. In this paper, we look at the effect of adding a channel both for the short term and over time. We call a period of time during which a transaction is served a *round*. In multiple rounds, transactions are repeated several times. We show that the paradox can appear in a single round if liquidity-based fees are used (Section III) or over multiple rounds if proportional fees are used (Section IV). An appearance of the paradox over multiple rounds means the addition of a channel (or similarly, the existence of a current channel) increases the total expected fee needed to be paid during all rounds. Our motivation for multiple payment rounds aims to allow users to perform a transaction more than once. Routing algorithms like Spider [14] suggest splitting a single transaction into multiple transactions that are sent to the network with a time delay; therefore, users may transmit similar transactions over several rounds. Moreover, recurring transactions (transactions with the same sender and receiver) are common in payment channel networks. The analysis in [13] of Ripple [31], [32] (an early proposed payment channel network) examined the percentage of recurring transactions. Over 1306 days of examined data, the vast majority of transactions (over 86%) were transmitted several times.

B. Paradox Impossibility Property

We start with a simple impossibility property.

Lemma 1. *In networks where the fees (weights) on the links are constant values, adding links can only decrease the*

payment cost at Nash Equilibrium; therefore, the single-round paradox cannot exist upon adding a link.

Proof. The proof relies on the fact that since the fees are constant, the route user i chooses does not affect the fee user j pays, and vice versa. In addition, as this is a single-round case, the fees on the links depend only on the initial state of the network at the beginning of the round. Hence, to reach a Nash Equilibrium, each user chooses its optimal route considering the parameters at the beginning of the round, regardless of the routes other users choose. Assuming there is only one user, adding a link does not affect the fees of the other links; therefore, it can only improve the optimal route of any user. Since each user chooses the optimal route regardless of what any other user chose, in case we have several users, adding the link can only improve the cost (i.e., decrease it) of a payment at Nash Equilibrium. $\square$

Property 1. *Consider channels using fees that are fixed or proportional to the transaction size but are insensitive to the channel liquidity. For a single round of payments, adding a new channel cannot increase the fees paid at Nash Equilibrium.*

We explain the intuition behind this property, which can be derived directly from a similar result of Dafermos et al. [33] for transportation networks. The meaning of the paradox in a single round is the addition of a channel that causes a fee increase during one round. Intuitively, if fees are not liquidity-based, the cost for a transaction using a channel is not influenced by whether another transaction uses this channel. Thus, a change in the path of a transaction to use the newly added channel is taken only if it improves its cost.

Beyond a single round of payments, given a transaction transmitted over the network for $k \geq 1$ rounds, we examine the total fee paid over all rounds. We define *Braess paradox in payment channel networks* as the total fee increase upon adding a new payment channel.

Property 1 states that Braess paradox cannot occur in payment channel networks using liquidity-independent fees in a single round. Though, Braess paradox might appear in payment channel networks and cause a fee increase.

C. Motivation for Liquidity-Based Fee Functions

We now explain the motivation for liquidity-based fees that encouraged us to analyze them in Section III and overview such existing fee functions.

A major issue with payment channel networks is channel imbalance [34]. A channel is limited to passing only transactions no larger than the channel's liquidity. If transactions use only one direction of the channel, its liquidity decreases fast and it becomes impossible to transmit any more transactions in this direction. The case where channels are imbalanced is very common, so usually, transactions probe several paths until one succeeds, which is an inefficient method [34], [35]. Since it is important to keep payment channels usable, several solutions are suggested. Routing algorithms for transactions such as Spider [14] and Flash [13] try to keep channels balanced.

To address imbalance, using liquidity-dependent fees has been suggested [28]–[30]. Assume there is a payment channel between users u_1 and u_2 , where u_1 locked a value of c_1 and u_2 locked a value of c_2 . Ren et al. [30] define channel imbalance I_{cap} as $I_{cap} = \frac{|c_1 - c_2|}{c_1 + c_2}$. If $I_{cap} = 0$ the bidirectional channel is equally balanced, and if $I_{cap} = 1$ the bidirectional channel is completely imbalanced. To optimize I_{cap} , they offer a fee function of the form $\Phi = 1/c_i^\beta$ where c_i is the channel's liquidity in the direction of the transaction and β is a positive tunable parameter.

Giovanni et al. [28] propose a different fee structure that is based on an alternative computation of the imbalance of the channel. Imbalance is defined as $|c_1 - c_2|$ and fees are determined based on the range value for the imbalance. Engelshoven and Roos [29] propose another fee function: For a bidirectional channel e between Alice and Bob, they define the reference point r as the liquidity values c_1 and c_2 that Alice and Bob consider optimal. While [28] considers the optimal reference point as one where the liquidity is split equally between the two directions, in practice, the channel is likely usable in one direction more than the other. Therefore, [29] suggests it might be more beneficial to consider an unequal distribution of the total liquidity. They propose two similar fee functions: For a transaction of size x that is transmitted from Alice's side to Bob's, and with a reference point r , the fee charged for transmitting the transaction is $\Phi = \alpha \cdot (1 + \frac{|c_1 - x - r| \cdot |c_1 - r|}{c_1 + c_2})$, where α is some normalization factor. If transmitting the transaction results in the channel getting farther from the reference point, then the fee function presented above is used. However, if transmitting the transaction gets the channel closer to the reference point, no fee is charged (i.e. $\Phi = 0$).

D. Lightning Specifications and their Routing Algorithms

The Basis of Lightning Technology (BOLT) [26] is a set of specifications that describe various protocols and standards of the Lightning network. The three main routing algorithms Lightning nodes implement are *C-lightning*, *LND* and *Eclair* [36]. These algorithms use source routing, preferring low fees and short-delay paths. Eclair uses Yen's algorithm [37] to find the n shortest paths according to the weight function it defines and then picks one path at random. In contrast, both *C-lightning* and *LND* use Dijkstra to find a (single) optimal path with weight functions described below. Thus, when examining the paradox potential in Lightning in Section V we use *C-lightning* and *LND* routing algorithms.

C-lightning weight function. For a payment channel e , this weight function is defined as $w(e) = (amt_e + scale \cdot fee_e) \cdot delay_e \cdot risk + bias$. Here $delay$ is the maximum time transmitting a transaction on the channel may take (otherwise it fails), fee is the fee the intermediary node charges and amt is the transaction size. $scale$ is a parameter uniformly sampled from $[1 - fuzz, 1 + fuzz]$, and $fuzz$, $risk$, $bias$ are tunable parameters with default values of 0.05, 10 and 1 respectively.

LND weight function. LND's weight function of payment channel e is defined as $w(e) = amt_e \cdot delay_e \cdot risk + fee_e + \frac{penalty}{prob_e}$. Here $delay$, amt and fee are the same as

for C-lightning, *penalty* and *risk* are tunable parameters with default values of 100 and $1.5 \cdot 10^{-8}$ respectively, and *prob* is a variable affected by the last time the channel failed to transmit a transaction. *prob* has a default value of 0.6 for channels without prior knowledge about them.

III. BRAESS PARADOX USING LIQUIDITY-BASED-FEES

A. Overview

We intend to show the possibility of the appearance of Braess paradox in payment channel networks. By Property 1, the paradox does not appear for a single round using only liquidity-independent fees. Hence, we first present the single-round paradox under the liquidity-based fee model and examine the paradox appearance conditions. We note that in a network with many participants, different users have different interests. In particular, a fee increase may even benefit some users. However, since payment channel networks aim to allow scalability of high transaction rates with low fees, it is in the network's interest to reduce fees. Throughout the paper, we view the paradox from the network's perspective, analyzing how the addition (or existence) of edges might increase the network fees.

For liquidity-based fees, the fee charged by a channel depends on the current liquidity of the channel. Under this fee model, when multiple transactions have to be served (at the same round), we assume all of their orders occur with equal probability. These orders change the updated liquidity of the channel when serving a transaction, so we refer to the *expected fees* across potential orders. Additionally, we assume some channels have high liquidity. Therefore, a transaction has a relatively small impact on the liquidity, making its fee approximately constant. Other channels with lower liquidity charge a fee that is liquidity-dependent. Our analysis of Lightning network data in Section V supports this assumption. For low-liquidity channels, we assume a liquidity-based fee function $\Theta(x) = \alpha \cdot \frac{x}{F-x}$ where x is the transaction's value and α is a normalization factor. $F \in \{c_1, c_2\}$ denotes the liquidity of the channel in the direction of the transaction.

For simplicity, we follow the common assumption in studies of the paradox, that the graph has symmetrical properties with two pairs of links having the same fee functions. Additionally, we assume there are only two users u_1 and u_2 , who both want to create a transaction from s_1 (start) to s_4 (end). Fig. 2 illustrates the liquidity-based graph, a graph such that links e_{13}, e_{24} have a fixed fee and links e_{12}, e_{34} refer to a fee based on the liquidity F . Although all links in Fig. 2 are unidirectional, in fact, the only channel we demand to be unidirectional in our analysis is e_{23} since making all other channels bidirectional does not add any new simple paths from s_1 to s_4 . This assumption fits the case where channels are disabled in one direction. Moreover, as shown in Section V, even when both directions are active, the fees in one direction may be much higher than in the other, and thus the channel may be used only in one direction in practice. This assumption is made to simplify the conditions presented later and can be removed in cases where the fee of the path $s_1 - s_3 - s_2 - s_4$ is higher than the fees of the other existing paths.

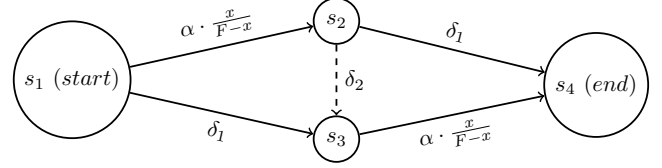


Fig. 2: (Liquidity-Based Graph) Illustration of Braess paradox in a payment channel network, serving transactions of value x from s_1 to s_4 . δ_1 and δ_2 represent a constant cost, F is the channel's liquidity and α is a constant normalization factor.

B. An Example of the Single-Round Paradox Using the Liquidity-Based Fee Model

To give an intuition for the paradox, we first present a particular example of it. We refer to the liquidity-based graph in Fig. 2, with the network parameters summarized in Table III.

TABLE III: Example for liquidity-based graph parameters

Symbol	δ_1, δ_2	α	F	x
Value	37, 20	90	14	2

Without link e_{23} , Nash Equilibrium is derived when each of the paths p_{up}, p_{down} is taken by one user so that each user pays a fee of $\delta_1 + \alpha \cdot \frac{x}{F-x} = 52$. If both users choose the same path, the expected fee each user pays is $\frac{1}{2} \cdot (90 \cdot \frac{2}{14-2} + 37) + \frac{1}{2} \cdot (90 \cdot \frac{2}{12-2} + 37) = 53.5$. Table IV shows the potential routing options when channel e_{23} does not exist.

TABLE IV: Potential paths and expected fees when the channel e_{23} does not exist. We neglect some of the possible paths that appear in the table due to symmetry.

Option	u_1 Path	u_2 Path	u_1 fee expectation	u_2 fee expectation
1	$s_1-s_2-s_4$	$s_1-s_2-s_4$	53.5	53.5
2	$s_1-s_3-s_4$	$s_1-s_3-s_4$	53.5	53.5
3	$s_1-s_2-s_4$	$s_1-s_3-s_4$	52	52
4	$s_1-s_3-s_4$	$s_1-s_2-s_4$	52	52

The paradox appears for instance upon adding a channel e_{23} with a constant fee of $\delta_2 = 20$. As summarized in Table V, the system reaches Nash Equilibrium when both users use $p_{mid} = start \rightarrow s_2 \rightarrow s_3 \rightarrow end$, implying a higher fee of 53 for each user. To conclude, adding e_{23} results in an increase in the fee each user needs to pay in Nash Equilibrium from 52 to 53.

TABLE V: Potential paths and expected fees upon the addition of channel e_{23} .

Option	u_1 Path	u_2 Path	u_1 fee expectation	u_2 fee expectation
1	$s_1-s_2-s_4$	$s_1-s_3-s_4$	52	52
2	$s_1-s_2-s_3-s_4$	$s_1-s_3-s_4$	51.5	53.5
3	$s_1-s_2-s_4$	$s_1-s_2-s_3-s_4$	53.5	51.5
4	$s_1-s_2-s_3-s_4$	$s_1-s_2-s_3-s_4$	53.0	53.0

Next, we show how the paradox can vanish with no outside intervention. The single-round paradox might appear in certain rounds and not appear in others. Upon serving two *additional* such transactions, using p_{mid} has a high fee of at least 68.75,

while paying 59.5 if users choose disjoint paths, as shown in Table VI. Therefore, the paradox lasts only for a single round of payments, and as proven later in Lemma 3, once the paradox vanishes, it will not appear again in later rounds.

In total, the paradox might appear in the network for a certain time and then vanish without any outside intervention.

TABLE VI: Potential paths and expected fees in serving a pair of transactions for the second time with the existence of the channel e_{23} .

Path Option	u_1 Path	u_2 Path	u_1 fee expectation	u_2 fee expectation
1	$s_1-s_2-s_4$	$s_1-s_3-s_4$	59.5	59.5
2	$s_1-s_2-s_3-s_4$	$s_1-s_3-s_4$	68.75	63.25
3	$s_1-s_2-s_3-s_4$	$s_1-s_2-s_3-s_4$	72.5	72.5
4	$s_1-s_3-s_4$	$s_1-s_2-s_3-s_4$	63.25	68.75

C. Conditions for the Paradox's Appearance in the Liquidity-Based Fee Model

We aim to find under what conditions opening a payment channel can increase the expected fee each user pays in Nash Equilibrium. The graph structure is presented in the liquidity-based graph (Fig. 2). We notice users may use one of three paths: $p_{up} = s_1-s_2-s_4$, $p_{down} = s_1-s_3-s_4$ or $p_{mid} = s_1-s_2-s_3-s_4$. For the paradox to appear, two conditions should hold:

(i) If u_1 and u_2 are at Nash Equilibrium, assuming e_{23} does not exist, after considering e_{23} it has to be worthwhile for at least one user to change their path. We note that by Lemma 2, if it is beneficial for one user to change path - it is worthwhile for the other user too.

(ii) The expected fee each user pays is higher than the fee paid if channel e_{23} did not exist.

Lemma 2. Under the liquidity-based graph presented in Fig. 2 and any liquidity-based fee function that gives constant values to e_{13}, e_{23}, e_{24} . Consider two users u_1, u_2 transmitting transactions with the same value. If it is beneficial for u_1 to use p_{mid} it is beneficial for u_2 to use p_{mid} too.

Proof. By the symmetry of the liquidity-based graph, the difference in cost in a transition from path p_{up} to p_{mid} equals the cost of a change from p_{down} to p_{mid} . It equals in both cases in expectation $\delta_2 + \frac{1}{2} \cdot \left(\alpha \cdot \frac{x}{F-x} + \alpha \cdot \frac{x}{F-2x} \right) - \delta_1$. $\square$

The first condition assures that the addition of e_{23} is not ineffective, meaning it is worth it for at least one user to use this link. The second condition assures that adding the link led to a fee increase, thus the paradox appeared.

Following the presented conditions, we examine the users' path options:

Case 1: In case channel e_{23} does not exist, as the graph is symmetrical, in Nash Equilibrium one user uses p_{up} and the other takes p_{down} . The fee each user pays is then $\Phi_{org} = A + \alpha \cdot \frac{x}{F-x}$.

Case 2: Following adding e_{23} , the expected fee u_1 pays if only she uses p_{mid} : $\Phi_{alone} = \alpha \cdot \frac{x}{F-x} + \delta_2 + \frac{1}{2} \cdot \left(\alpha \cdot \frac{x}{F-x} + \alpha \cdot \frac{x}{F-2x} \right)$. It uses solely one of the links e_{12} or e_{34} and shares the other link with the second transaction so it can be the first or second to use the other link.

Case 3: The expected fee each user pays if both u_1 and u_2 use p_{mid} : $\Phi_{same} = \alpha \cdot \frac{x}{F-x} + \delta_2 + \alpha \cdot \frac{x}{F-2x}$.

The first case presents the fee each user pays in Nash Equilibrium state when e_{23} does not exist. The second case presents the fee where e_{23} exists and one user uses it alone, and the third case presents the case both users choose the same path and use e_{23} .

If the first condition holds, i.e. $\Phi_{org} \leq \Phi_{same}$, it is worthwhile for one user to change paths and use p_{mid} , and due to Lemma 2, Nash Equilibrium is received when both users use p_{mid} . If in addition the second condition holds, meaning $\Phi_{org} \leq \Phi_{same}$, the fee each user pays after the addition of e_{23} increases and we observe the paradox.

By the two inequalities, we observe the following:

Theorem 1. The single-round paradox appears in the liquidity-based graph when

$$\frac{1}{2} \cdot \left(\frac{1}{F/x - 1} + \frac{1}{F/x - 2} \right) \leq \frac{\delta_1 - \delta_2}{\alpha} \leq \frac{1}{F/x - 2}.$$

By these conditions, we derive the following intuitive points: (i) The dependency in δ_1, δ_2 is through their difference ($\delta_1 - \delta_2$). For large enough δ_2 it might not be worth using the new channel e_{23} . On the other hand, for large δ_1 , it is beneficial for both users to use the path p_{mid} and not the channels with cost δ_1 . (ii) Similarly, the dependency in F and x is through their ratio F/x .

D. Appearance and Disappearance of the Paradox in the Liquidity-based Fee Model

An interesting property of Braess paradox in payment channel networks is the appearance and disappearance of the paradox without any outside intervention.

Lemma 3. Under the liquidity-based fee model and the liquidity-based graph, p_{mid} eventually becomes either suboptimal (i.e., more expensive than alternative paths) or unusable due to the unavailability of liquidity. Moreover, once this happens, it does not change back for any later round.

Proof. The fees of e_{12} and e_{34} increase over time and are not bounded from above while fees on the other links do not change. Since unlike paths p_{up} and p_{down} , p_{mid} makes use of both e_{12} and e_{34} , it eventually reaches a higher cost than p_{up} and p_{down} unless it becomes unavailable earlier due to its bounded liquidity. $\square$

Lemma 4. Consider the liquidity-based fee model and the liquidity-based graph. If the paradox appears in a certain round and vanishes in a later round then it does not appear again afterward.

Proof. There are two possibilities for the paradox not to appear: Either link e_{23} is so cheap that using p_{mid} decreases the fee, or it is not worth using e_{23} since p_{mid} is more expensive. Since the paradox did appear at some round, necessarily using p_{mid} did decrease the fee at that round, and later became unworthy to use. By Lemma 3 p_{mid} will not become worthy again in later rounds. $\square$

Additionally, under the above model, assuming the initial liquidity of e_{12} and e_{34} is F_0 and each user transmits a transaction of size x over the network, we calculate the number of rounds k until the paradox may appear. The paradox appears if $\Phi_{alone} \leq \Phi_{org}$, and $\Phi_{org} \leq \Phi_{same}$. If $\Phi_{alone} > \Phi_{org}$, p_{mid} is unworthy to use and by Lemma 3 p_{mid} stays unworthy to use hence the paradox will not appear later. If the condition $\Phi_{same} \leq \Phi_{org}$ is met, both users change their strategy so that they use channel e_{23} . Therefore, the liquidity of e_{12} and e_{34} over time is $F(k) = F_0 - 2 \cdot k \cdot x$.

Recall $\Phi_{org} = \delta_1 + \alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - x}$ and $\Phi_{same} = \alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - x} + \delta_2 + \alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - 2x}$. We aim to find the minimal k for which: $\Phi_{same} > \Phi_{org}$. From this inequality, we express the smallest k that maintains it as $k_{appear} = \left\lfloor \frac{F_0}{2x} - 1 - \frac{\alpha}{2 \cdot (\delta_1 - \delta_2)} \right\rfloor$.

We also want to find out how long the paradox lasts. Since the fees on channels e_{12} and e_{34} only get higher over time while the fees of e_{13} and e_{24} do not change, eventually $\Phi_{alone} > \Phi_{org}$ and it will not be worth using e_{23} for either user. Recall $\Phi_{alone} = \frac{1}{2} \cdot (2 \cdot \alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - x} + \delta_2) + \frac{1}{2} \cdot (\alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - x} + \delta_2 + \alpha \cdot \frac{x}{F_0 - 2 \cdot k \cdot x - 2x})$, we find the smallest k s.t. $\Phi_{alone} > \Phi_{org}$. This implies:

$$k_{vanish} = \left\lfloor \frac{-\beta + \sqrt{\beta^2 - 4 \cdot \gamma \cdot \varphi}}{2} \right\rfloor,$$

such that $\gamma = 4 \cdot \frac{2 \cdot (\delta_1 - \delta_2)}{\alpha \cdot x} \cdot x^2$, $\beta = 4 \cdot x + \frac{2 \cdot (\delta_1 - \delta_2)}{\alpha \cdot x} \cdot [-4 \cdot x \cdot F_0 + 6x^2 - 4 \cdot x]$ and $\varphi = 3x - 2 \cdot F_0 + \frac{2 \cdot (\delta_1 - \delta_2)}{\alpha \cdot x} \cdot [F_0^2 + 3 \cdot x \cdot F_0 + 2 \cdot F_0]$.

The intuition is as follows. The above implies an equality $\gamma \cdot k^2 + \beta \cdot k + \varphi \leq 0$. γ is a positive number therefore our function is a minimum parabola. Since k can only be positive too, for every k that sustains $0 \leq k \leq \left\lfloor \frac{-\beta + \sqrt{\beta^2 - 4 \cdot \gamma \cdot \varphi}}{2} \right\rfloor$ the paradox will appear.

Note that if $\Phi_{alone} > \Phi_{org}$ for k_{appear} , the paradox does not appear as it misses its appearance interval.

IV. BRAESS PARADOX USING PROPORTIONAL FEES

In this section, we provide an example and analyze the conditions for the paradox existence in payment channel networks using proportional fees. For simplicity reasons, at first, we assume all links on the graph are unidirectional, though this assumption will be removed in Section IV-E. Recall payment channel networks use fee functions of the form $\Theta(x) = \gamma \cdot x + \delta$ where $\gamma \cdot x$ is a proportional fee dependent on the transaction value x and δ is a base fee. As presented in Section II-D, existing routing methods in Lightning and Raiden take into account only the current round and try to find the cheapest path usable at this round (acting greedily), hence we assume users use a greedy algorithm as well.

A. Braess Paradox Example Using Proportional Fees

Consider the graph presented in Fig. 3 where transactions are sent from s_1 (start) to s_4 (end). $\gamma_A, \gamma_B, \gamma_D, \gamma_E$ represent a fee function of the form $\Theta(x) = \gamma \cdot x$ (here the base fee δ has a low value and practically equals 0). Values F_A, F_B, F_D, F_E represent the initial channel liquidity. In this example, we assume the graph parameters are: $\gamma_A = 1, \gamma_B = 3, \gamma_D =$

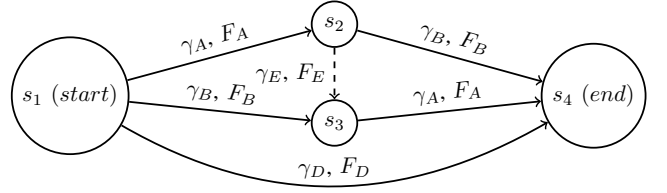


Fig. 3: Braess paradox for proportional fee functions. $\gamma_A, \gamma_B, \gamma_D, \gamma_E$ represent a fee function of the form $\Theta(x) = \gamma \cdot x$. The values F_A, F_B, F_D, F_E represent the initial channel liquidity.

$\gamma, \gamma_E = 0, F_A = 1, F_B = F_D = F_E = 2$. We look at the case where $k = 2$ and $x = 1$. We notice transactions may take one of four paths: $p_{up} = s_1-s_2-s_4$, $p_{down} = s_1-s_3-s_4$, $p_{mid} = s_1-s_2-s_3-s_4$ or $p_{other} = s_1-s_4$.

The paradox relies on the fact that opening a payment channel may result in a load on certain channels, potentially making them unusable later. Although under this fee model liquidity has no direct impact on the fee of a channel, it determines its availability, potentially enforcing the use of channels with high fees.

Recall e_{ij} denotes the channel connecting s_i and s_j . If e_{23} does not exist, at each round, user u uses either p_{up} or p_{down} , paying a fee of $2 \cdot (1 + 3) = 8$. But if e_{23} does exist, for the first round u uses p_{mid} , but then p_{mid}, p_{up} and p_{down} become unusable; hence, u must use p_{other} for the second round. In total, u pays a total fee of $1 \cdot (1 + 1) + 1 \cdot 7 = 9$, a fee increase.

To conclude, opening a new payment channel increased the fee each user needed to pay over the two rounds from 8 to 9.

B. Conditions for the Paradox Appearance Under The Proportional Fees Model

After showing that the paradox can exist using proportional fee functions, we provide a simple condition.

Theorem 2. Consider a graph in the form of Fig. 3 with one user u transmitting a transaction of size x for $k > \frac{2 \cdot F_A}{x}$ rounds. Assuming $F_A < F_B$, adding link e_{23} with liquidity F_E and fee γ_E such that $F_E > 2 \cdot F_A$ and $2 \cdot \gamma_B - \gamma_D < \gamma_E < \gamma_B - \gamma_A$ will cause the paradox to appear.

Proof. For simplicity reasons only (this assumption is not mandatory), we assume F_A is divided by x with no remainder. If e_{23} does not exist, for $\frac{F_A}{x}$ rounds u uses p_{up} , for $\frac{F_A}{x}$ more rounds she uses p_{down} , and for the left rounds she uses p_{other} . In total, the fee paid before p_{up} and p_{down} become unusable is $2 \cdot \frac{F_A}{x} \cdot (\gamma_A + \gamma_B) \cdot x = 2 \cdot F_A \cdot (\gamma_A + \gamma_B)$.

Taking e_{23} into account, if u wants to use p_{mid} she can do so for only $\frac{F_A}{x}$ rounds, afterward she must use p_{other} . The total fee paid for the first $2 \cdot \frac{F_A}{x}$ rounds is $(\frac{F_A}{x} \cdot (\gamma_A + \gamma_E + \gamma_A) + \frac{F_A}{x} \cdot \gamma_D) \cdot x = F_A \cdot (\gamma_A + \gamma_E + \gamma_A) + F_A \cdot \gamma_D = F_A \cdot (2 \cdot \gamma_A + \gamma_E + \gamma_D)$.

We notice that whether e_{23} exists or not, after at most $2 \cdot \frac{F_A}{x}$ rounds u must use p_{other} , hence when comparing between fees we only look at the first $2 \cdot \frac{F_A}{x}$ rounds. When adding e_{23} , the paradox appears if two conditions are met:

(i) Adding a new link creates a path with a lower fee than other optional paths, i.e. $(\gamma_A + \gamma_E) \cdot x < \gamma_B \cdot x$.

(ii) The total fee paid when e_{23} exists is higher, i.e. $2 \cdot F_A \cdot (\gamma_A + \gamma_B) < F_A \cdot (2 \cdot \gamma_A + \gamma_E + \gamma_D)$.

From the first inequality we derive the inequality $\gamma_E < \gamma_B - \gamma_A$, and from the second inequality we obtain the formula $2 \cdot \gamma_B - \gamma_D < \gamma_E$.

To summarize, adding a new link e_{23} causes a fee increase if $2 \cdot \gamma_B - \gamma_D < \gamma_E < \gamma_B - \gamma_A$. $\square$

Corollary 1. *When using fixed fee functions, i.e., of the form $\Theta(x) = \gamma \cdot x + \delta$ where γ equals zero, the conditions presented in Theorem 2 apply as well.*

Corollary 1 stems from the fact that x is reduced from our equations and does not affect the conditions presented above, thus they are applicable in a graph with fixed fees.

C. The Paradox in Asymmetrical Graphs Using Proportional Fees

We now generalize the previous analysis from the symmetrical case to a more general graph structure, as illustrated in Fig. 4. The same graph structure as before is examined, but now with no assumption on symmetry among channels. The fee functions have the general form of $\Theta(x) = \gamma \cdot x + \delta$. We simply refer to such graphs as asymmetrical.

If e_{23} does not exist (or when it is not worth using it), p_{up} is usable for $\frac{\min\{F_1, F_2\}}{x}$ rounds and p_{down} for $\frac{\min\{F_3, F_4\}}{x}$ rounds. For the sake of simplicity, we assume $F_1 \leq F_2$ and $F_1 \leq F_4$ as well as $F_4 \leq F_3$ and that all these values are divisible by x . These assumptions can easily be generalized.

Assume $k > F_1 + F_4$. For the paradox to appear, p_{mid} must be the path with the lowest fee so the addition of e_{23} will not be ineffective, that is $\Phi_{mid} < \min\{\Phi_{up}, \Phi_{down}\}$, where:

$$\Phi_{mid} = \Theta_1(x) + \Theta_5(x) + \Theta_4(x),$$

$$\Phi_{up} = \Theta_1(x) + \Theta_2(x), \Phi_{down} = \Theta_3(x) + \Theta_4(x).$$

Before e_{23} existed the total fee needed to be paid was:

$$\Phi_{without e_{23}} = \Phi_{up} \cdot \frac{F_1}{x} + \Phi_{down} \cdot \frac{F_4}{x} + \Theta_{other}(x) \cdot \left(k - \frac{F_4}{x} - \frac{F_1}{x}\right).$$

When adding e_{23} , p_{mid} is the cheapest path hence u uses it for $\frac{\min\{F_1, F_4\}}{x}$ rounds with a fee of Φ_{mid} at each round. As $\min\{F_1, F_4\} = F_1$, when p_{mid} becomes unusable, u uses p_{down} for $F_4 - F_1$ additional rounds, paying a fee of Φ_{down} at each round. For the $k - \frac{F_4}{x}$ rounds left, only p_{other} is usable, with a fee of $\Phi_{other}(x)$ at each round. Hence, the total fee paid is

$$\Phi_{with e_{23}} = \frac{F_1}{x} \cdot \Phi_{mid} + \frac{F_4 - F_1}{x} \cdot \Phi_{down} + \left(k - \frac{F_4}{x}\right) \cdot \Phi_{other}.$$

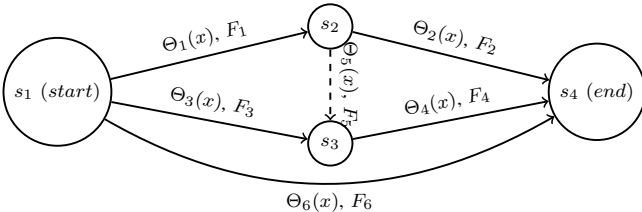


Fig. 4: Braess paradox in an asymmetrical graph structure. $\Theta_i(x), F_i$ refer to the fee function and liquidity in channel i .

The paradox expresses an increased fee upon the addition of e_{23} . Thus for the paradox to appear in the asymmetrical case with proportional fees, the following conditions must hold:

- (i) $\Phi_{mid} < \min\{\Phi_{up}, \Phi_{down}\}$
- (ii) $\Phi_{with e_{23}} > \Phi_{without e_{23}}$

D. Path-to-Channel Reduction For Proportional Fees

We now present a reduction from a more general graph, the *Three-Path graph* presented in Fig. 5, to the *Path-Reduction graph* presented in Fig. 6. This allows us to apply the conditions presented in Section IV-C on the Three-Path graph.

Given a path p , we define the *path's fee* as the sum of fees over p , as this is the total fee needed to be paid when using p . Additionally, we define the *path's liquidity* as the minimal liquidity of a channel on the path.

Observation 1: The fee function of path p is of the form $\gamma \cdot x + \delta$ too (and equals the sum of the fees over the links of the path).

Observation 2: The total amount of funds a path can serve equals the liquidity of the minimal-liquidity channel on it.

After defining a path's fee and liquidity, we present the following reduction: Assume we add a channel between node s_{1i} on p_1 and node s_{2j} on p_2 . Denote s_{1i} as s' and s_{2j} as s'' . Now, we create a new graph similar to Fig. 4, where: Path $start - s_{12} - \dots - s_{1i}$ is represented by channel $start - s'$, path $s_{1i} - \dots - end$ is represented by channel $s' - end$, path $start - s_{22} - \dots - s_{2j}$ is represented by channel $start - s''$, path $s_{2j} - \dots - end$ is represented by channel $s'' - end$ and path $start - s_{32} - \dots - end$ is represented by channel $start - end$, as presented in Fig. 6.

Theorem 3. *Given a Three-Path graph as presented in Fig. 5, when adding channel e between a node s_{1i} on p_1 and a node s_{2j} on p_2 , we can create a Path-Reduction graph. Under the assumptions presented in Section IV-C, if the paradox appears in the Path-Reduction graph it appears in the Three-Path graph too.*

Proof. Intuitively, by Observation 2 channels in the Path-Reduction graph are usable as long as the paths in the Three-Path graph are usable. Since, by Observation 1, the fees in the Path-Reduction graph are of the same form and equal to those in the Three-Path graph, the two graphs are fee-equivalent. Therefore, if adding a channel implies the paradox in the Path-Reduction graph, it also causes the paradox to appear in the Three-Path graph.

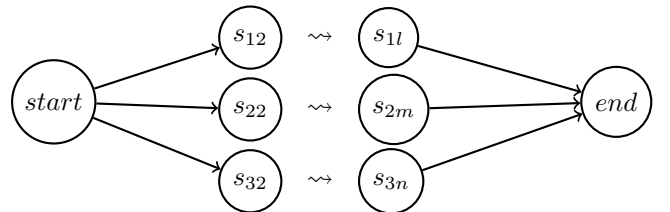


Fig. 5: The Three-Path graph, containing three disjoint paths p_1, p_2, p_3 from $start$ to end . The node name e_{ij} symbolizes this is the j 'th node on path p_i .

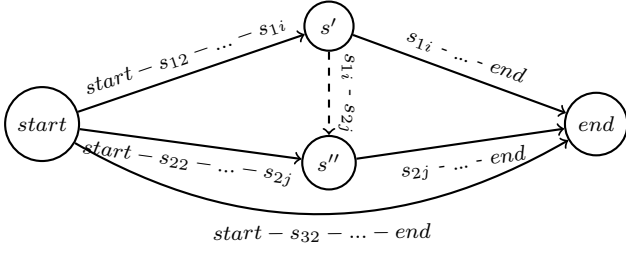


Fig. 6: The Path-Reduction graph. In this graph, each channel represents the path that appears on it. The liquidity and fee of each channel are the liquidity and fee of the path it represents, as defined in Section IV-D.

Formally, assume the paradox holds for the Path-Reduction graph, that is in the graph presented in Fig. 6 too. By Observation 1, the fees over p_{up} , p_{mid} , p_{down} , p_{other} in Fig. 4 are similar to the fees needed to be paid on the corresponding paths in Fig. 5. On Fig. 4, before adding e p_{up} can be used for k_{up} rounds, and p_{down} for k_{down} rounds. In total, the fee needed to be paid for $k = k_{up} + k_{down}$ rounds is $fee_{before} = (\Phi_{p_{up}} \cdot k_{up} + \Phi_{p_{down}} \cdot k_{down})$. By Observation 2, we infer that in Fig. 5 the corresponding paths to p_{up} and p_{down} can be used for k_{up} , k_{down} rounds too, respectively. Since the fees in Fig. 5 are the same as Fig. 4, the total fee needed to be paid for k rounds equals fee_{before} . The paradox appears in Fig. 4, hence by adding e the cost of p_{mid} is lower than $\min\{p_{down}, p_{up}\}$, therefore it is worth using. In addition, the total fee needed to be paid is $fee_{after} = \Phi_{p_{mid}} \cdot k_{up} + \Phi_{p_{down}} \cdot (k_{down} - k_{up}) + \Phi_{other} \cdot k_{up}$. Moreover, $fee_{after} > fee_{before}$. By Observation 2, in Fig. 5 p_{mid} is usable for k_{up} rounds too, hence p_{mid} will be usable for $k_{down} - k_{up}$ more rounds, and we will need to use p_{other} for the last k_{up} rounds. The total fee paid after adding e in Fig. 5 equals fee_{after} . In total, in Fig. 5 $fee_{after} > fee_{before}$, therefore the paradox appears in Fig. 5 too. $\square$

The Path-Reduction graph maintains the attributes of the asymmetrical graph presented in Fig. 4. Hence, the conditions presented in Section IV-C apply to it. By Theorem 3, given a Three-Path graph, a Path-Reduction graph can be created to test the paradox appearance. Given a graph with more than three disjoint paths, a heuristic approach is to consider the three cheapest paths, ignoring other paths.

E. Conditions for the Paradox Appearance with Bidirectional Links

So far in this section, for simplicity, we assumed all links in the network were unidirectional. In practice, since besides e_{23} converting links to be bidirectional does not add any new paths between $start$ and end , the only link we demanded to be unidirectional in our analysis was e_{23} . Towards presenting modified conditions for the case where e_{23} (and all other graph links) is bidirectional, we start by presenting a useful property.

Property 2. Assuming there are enough paths (and liquidity) for performing transactions successfully at all rounds (transactions always succeed), and there is more than one path

disjoint to p_{up} and p_{down} (i.e. more than one path that can be used as p_{other}). In such a case, we can refer to the cheapest path among them as p_{other} and ignore all other paths when using the conditions presented in Theorem 2 and Section IV-C.

Intuitively, since in the equations of Theorem 2 and Section IV-C we use a value of p_{other} fee that might be lower than the actual fee paid when e_{23} exists, the fee calculated in the equations is lower (or equal) than the actual fee needed to be paid. If the condition of Theorem 2 and Section IV-C still hold and the calculated fee after adding e_{23} is an increase compared to the fee paid when e_{23} did not exist, then the actual fee paid after adding e_{23} is necessarily higher than the fee paid before e_{23} existed, hence the addition of the new link caused a paradox appearance.

More formally, recall $\Phi_{without e_{23}}$ is the fee paid when e_{23} does not exist and p_{other} is the cheapest path available after p_{up} , p_{down} and p_{mid} become unusable. Denote $\Phi_{with e_{23}}$ as the fee needed to be paid if e_{23} does exist, assuming when p_{up} , p_{down} and p_{mid} become unusable the fee the user pays at each round equals the fee of p_{other} (even if in practice p_{other} becomes unusable and another path is used). Finally, denote Φ_{actual} the actual fee needed to be paid if e_{23} exists.

Property 2 is based on the observation that since $\Phi_{actual} \geq \Phi_{with e_{23}}$, if the equations in Theorem 2 and Section IV-C hold then $\Phi_{with e_{23}} > \Phi_{without e_{23}}$, thus $\Phi_{actual} > \Phi_{without e_{23}}$ and the addition of e_{23} causes a fee increase. We are now ready to show the following property.

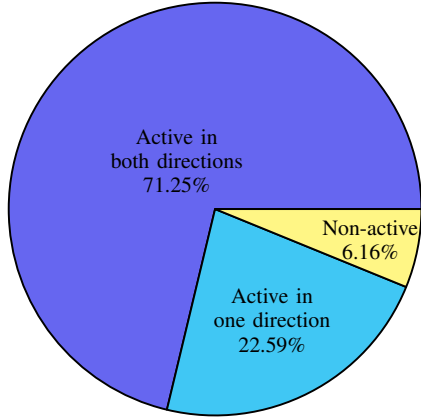
Theorem 4. Assuming routing only via simple paths (i.e., with no cycles) and nonnegative fees, the conditions presented in Theorem 2 and in Section IV-C hold when links are bidirectional.

Proof. Recall that all transactions are from $start$ to end . Since only simple paths are in use, turning all links bidirectional adds only one path that did not previously exist: $p_{bidirectional} = start \rightarrow s_3 \rightarrow s_2 \rightarrow end$. We divide into cases and observe that in all cases if either of the conditions in Theorem 2 or Section IV-C are met, the fee of $p_{bidirectional}$ is higher than p_{up} , p_{down} and p_{mid} as:

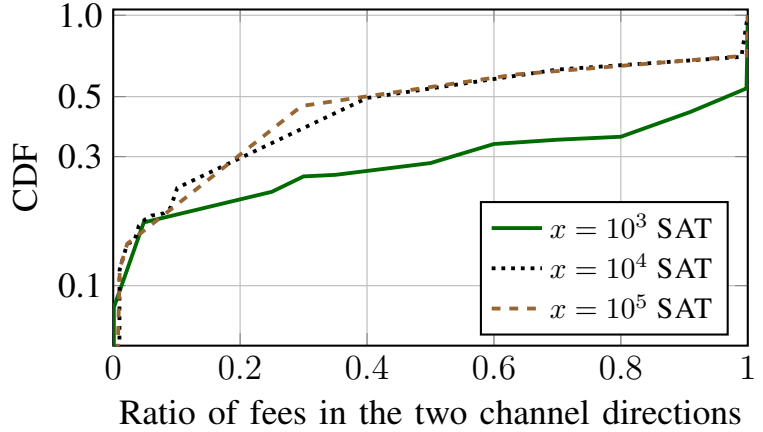
(i) *Symmetric graph model* - Denote the fee of $s_3 \rightarrow s_2$ as γ'_E . If $\gamma_A + \gamma_E < \gamma_B$, since $\gamma_E \geq 0$ necessarily $\gamma_A < \gamma_B$. Since $\gamma'_E \geq 0$ too, $2 \cdot \gamma_B + \gamma'_E > 2 \cdot \gamma_A$ and $2 \cdot \gamma_B + \gamma'_E > \gamma_A + \gamma_B$, hence the fee of $p_{bidirectional}$ is higher than p_{up} , p_{down} and p_{mid} .

(ii) *Asymmetric graph model* - Denote the fee of $s_3 \rightarrow s_2$ as Θ'_5 . If $\Phi_{mid} < \min\{\Phi_{up}, \Phi_{down}\}$, since $\Theta_5 \geq 0$ necessarily $\Theta_1 + \Theta_4 < \Theta_1 + \Theta_2$ hence $\Theta_4 < \Theta_2$, and $\Theta_1 + \Theta_4 < \Theta_3 + \Theta_4$ hence $\Theta_1 < \Theta_3$. Since $\Theta'_5 \geq 0$, necessarily $\Theta_1 + \Theta_2 = \Phi_{up} < \Phi_{bidirectional} = \Theta_2 + \Theta_3 + \Theta'_5$, $\Phi_{down} = \Theta_3 + \Theta_4 < \Phi_{bidirectional} = \Theta_2 + \Theta_3 + \Theta'_5$ and $\Phi_{mid} < \min\{\Phi_{up}, \Phi_{down}\} < \Phi_{bidirectional}$.

Our observation assures that if the conditions presented in Theorem 2 and Section IV-C hold, $p_{bidirectional}$ is not used until p_{up} , p_{down} and p_{mid} become unusable. Additionally, we notice that when using $p_{bidirectional}$ we do not add any liquidity to p_{up} , p_{down} or p_{mid} , hence they do not become usable again and $p_{bidirectional}$ can be treated as another expensive



(a)



(b)

Fig. 7: (a) Activity distribution of Lightning channels as of March 2020 (with 22460 channels connecting 2844 nodes). Channels may be in one of three states: Active in both directions, active in one direction and non-active (disabled in both directions). (b) CDF of fees ratio in the two directions in channels active in both directions, assuming transactions of size 10^3 , 10^4 and 10^5 SAT. The ratio for a channel is calculated as the lower fee divided by the higher fee among the fees in the two channel directions.

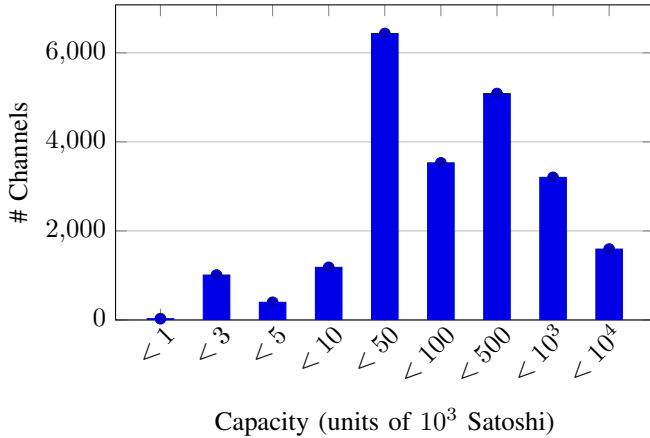


Fig. 8: Amount of channels vs. capacity size. The total number of channels was 22460.

path, just as p_{other} . In case the fee of $p_{bidirectional}$ is higher than p_{other} by Property 2 we can ignore it. Otherwise, we ignore p_{other} and use $\Theta_6 = \Phi_{bidirectional} = \Theta_2 + \Theta_3 + \Theta'_5$. $\square$

V. INDICATIONS FOR THE PARADOX IN THE LIGHTNING NETWORK

We refer to Braess paradox in payment channel networks as the occurrence of a fee increase following the establishment of a new payment channel. Previous sections analyzed the theoretical conditions for the appearance of Braess paradox in layer-2 payment networks. The paradox was analyzed on the classic Braess-paradox graph. In reality, a blockchain network is composed of a more complex graph with other nodes and edges. However, since less than 4% of the payments use a path with more than three intermediate nodes [38] and since Section IV-C generalizes the paradox graph, we wish to show

the effect of Braess paradox in real-life data. In this section, we study the Lightning network, Bitcoin's layer-2 payment channel network to provide indications for the appearance of the paradox in real data. We refer to Lightning data from March 2020, originally collected and presented in [38]. The network had a total of 2844 nodes and 22460 channels.

A. Lightning Network Channel Status and Fees

In Section II-A, we claim that although opening a channel between users may allow transfers in both directions, in practice, many Lightning channels might be unidirectional, as shown in Fig. 7. Fig. 7a presents the activity distribution of Lightning channels, as channels might be disabled in one or both directions. Almost 23% of Lightning channels are active only in one direction. Moreover, Fig. 7b presents the CDF (cumulative distribution function) of the fee ratio between the lower fee direction and the higher fee direction for channels active in both directions, using transactions of size 10^3 , 10^4 and 10^5 SAT (Satoshi = 10^{-8} BTC, 10^4 SAT $\approx$ \$2 as of October 2022) to calculate fees (recall fees depend on the transaction size). Although in many channels the fees are relatively similar (e.g. for transactions of size $x = 10^3$ 47% of the channels have a fee ratio of 0.99); however, in many channels the fee difference between directions was significant: For transactions of size 10^4 and 10^5 SAT, more than 10% of the channels had a fee ratio of 0.01 or lower, meaning the fee in one direction was more than 100 times higher than the fee in the opposite direction. For transactions of 10^3 SAT, more than 20% of channels had a fee ratio lower than 0.01. Since the differences between the base fees were more significant than the proportional fees, for smaller transactions the fee ratio between sides was even higher. This shows that although 71% of Lightning channels are active in both directions, in

practice many of these channels are probably used mainly in one direction.

Moreover, in Section III in the liquidity-dependent fee model, we assume some channels have high liquidity (hence their fee is approximately constant) while other channels have low liquidity (and charge a liquidity-dependent fee). We observe Lightning network has a high variance in the liquidity of channels that supports our assumption on channels with liquidity-based fees along with channels with fixed fees. Fig. 8 shows the distribution of the (total) capacity in the various channels. While 9051 channels have a capacity of at most $50 \cdot 10^3$ SAT, 4795 of the channels have a capacity of at least $500 \cdot 10^3$ SAT which is at least $\times 10$ higher. Such channels of low capacity might become unusable as assumed for some channels in part of the illustrations for the paradox.

Finally, we note that Lightning is a growing network, in which the addition of new payment channels is relatively frequent. Previous analysis [21] showed that the average node degree in the network increased from 1.71 in July 2018 to 14.38 by February 2019, indicating the network growth and edge addition. Additionally, the network increased from almost 2900 nodes and 22500 channels in March 2020 to almost 17000 nodes and more than 86000 channels by June 2025¹. Moreover, as payment channels have a time expiry, the addition of new channels is an integral part of the network.

B. Indications for the Multi-Round Paradox in Lightning Network

We now provide indications for the paradox appearance in Lightning network, where proportional fees are used. Since the liquidity balance of each user in a specific channel is private [39] and only information on the total locked liquidity in the channel is available, during the experiment, we used several liquidity distribution assumptions over the channels.

In the experiment, two random nodes were selected, identifying the three available lowest-weight paths: p_{up} , p_{down} (as the two lowest) and p_{other} (as the third). Given transaction t of size x (between the two nodes), we calculated the maximal number of rounds k in which p_{up} and p_{down} are usable. Then, the fee that would have been paid if t was transmitted for k rounds was evaluated. Finally, we tested if it is possible to add channel e that connects a node from p_{up} to a node from p_{down} so the fee for transmitting t for at least k rounds would increase. By creating a path connecting the two cheapest paths, the liquidity on p_{up} and p_{down} might run out faster when e exists, forcing t to use a more expensive path p_{other} . A large number of nodes with no repeats were randomly selected to examine the paradox frequency. Since the Lightning graph contains a large number of nodes and channels, the experiment was repeated with a restriction on path lengths of 2, 3 and 4 channels. Performing the analysis with path length limitations of 2 and 3, there were 150000 drawn source and destination node pairs. As the evaluation on paths of length 4 was computationally intensive, we drew only 5000 such pairs of nodes and analyzed the chances for the paradox to appear.

The analysis was performed using three different routing algorithms: Lowest-fee, C-lightning, and LND. All three algorithms find the lowest-weight paths for some particular weight function. Given a transaction of size x , in the lowest-fee algorithm, the weight of a channel is the fee a transaction needs to pay when using it. C-lightning and LND weight functions were defined in Section II-D. The algorithms' default values were used, assuming no previous knowledge of the links, meaning *probe* had a default value too. We notice the only channel-related parameters that affect the weight of each channel are the *TLD* value (time lock delta, the minimum number of blocks a node requires to be added to the expiry of HTLC [4]) which affects *delay*, and the fee each channel charges. First, to understand the potential of the paradox appearing in the network, we examined the impact of adding channel e with the lowest weight found: The lowest fee and TLD values found in the network were 0 and 4, respectively. As shown in Fig. 9, when limiting paths to two channels at most, the paradox does not appear in more than 0.1% of the cases for any transaction size. When allowing longer paths, the paradox frequency increases and reaches almost 20% with a transaction of size $15 \cdot 10^4$ SAT. We also observe that when assuming the network channels have more liquidity in the direction we use, the chances for the paradox to appear are higher. Additionally, in most cases for large transactions the paradox frequency was low, since the number of times p_{up} , p_{down} were usable was low and p_{other} was used for fewer rounds, having a lower effect on fees. From here on, all experiments were run with a path length limit of three channels, since when using this limitation the paradox was frequent, and we were able to analyze the experiment with a large number of source and destination nodes drawn.

The fee increase caused by the paradox and the number of rounds until the addition of e becomes unworthy were examined too. As shown in Fig. 10, when adding e causes a paradox appearance, the total fee increase (over all rounds) may reach almost 300% of the fee paid if we did not add e . This happens since the network is composed of many cheap but low-liquidity edges, which, when they run out of liquidity, leave only expensive channels to use.

Additionally, for non-small transactions, it does not take a large number of rounds k until the addition of e becomes unworthy, as for transactions of size $5 \cdot 10^4$ SAT, on average among the cases of detected fee increase, the increase was encountered within 5 rounds.

The paradox appears when using C-lightning and LND routing algorithms too. As presented in Fig. 11, the frequency of the paradox when using LND and C-lightning is similar to the lowest-fee algorithm, being able to cause a fee increase of almost 30% for small transactions. Interestingly, for larger transactions, the paradox appearance percentage using LND decreases. We assume this happens because for large transactions fees have high significance on LND's weight function (compared to other parameters). Hence, p_{mid} is used only when it is significantly cheaper, and using it does not cause a fee increase as much as small transactions do.

¹<https://mempool.space/lightning>

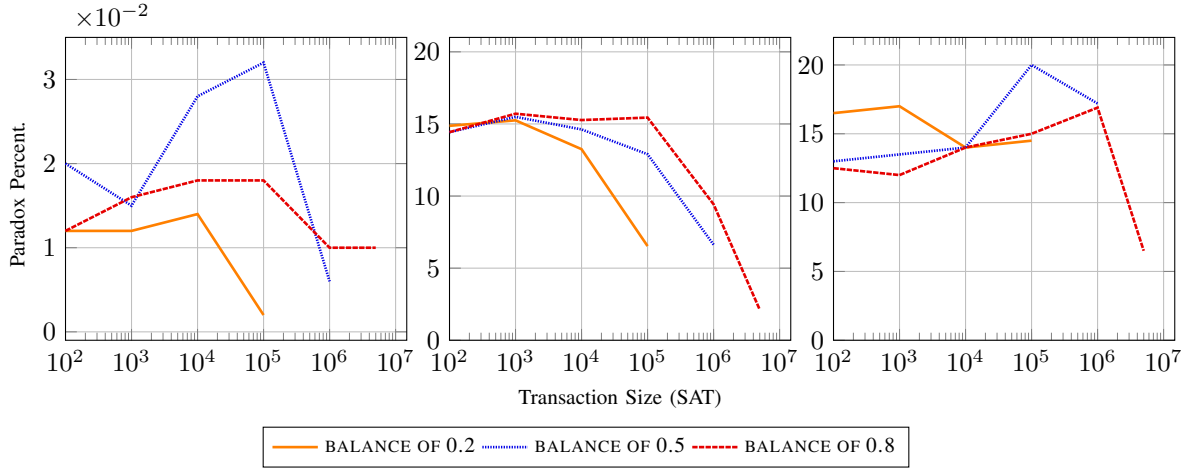


Fig. 9: Percentage of times the paradox is observed upon adding a free-fee channel using the lowest-fee algorithm, for path limitations of 2 (left), 3 (mid) and 4 (right) channels per path.

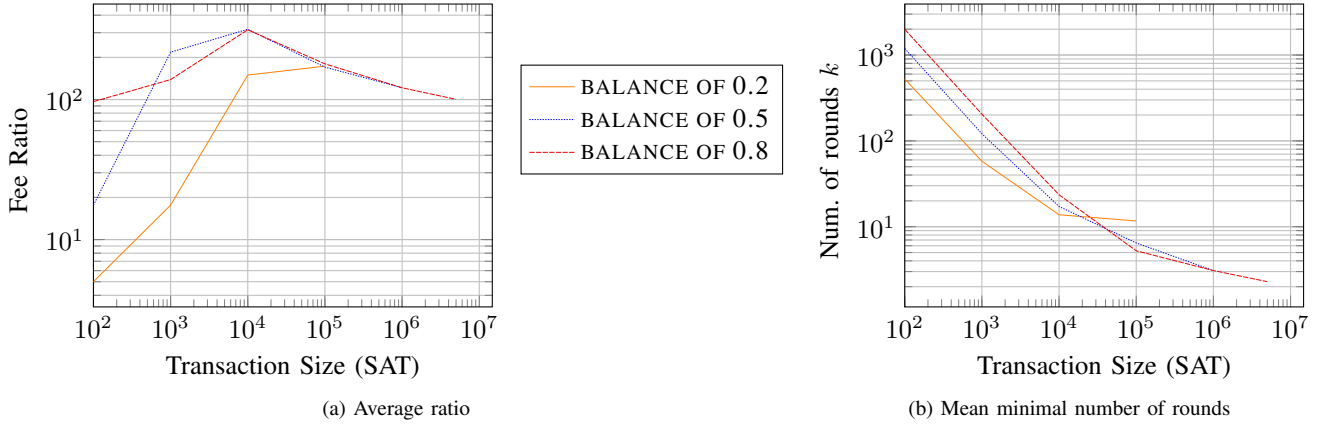


Fig. 10: (a) Average ratio between fees after and before adding the channel when the paradox appeared. (b) Mean minimal number of rounds for the paradox to appear. Channels are weighted by fees (lowest-fee). Curves refer to various distributions of the channel liquidity among its directions

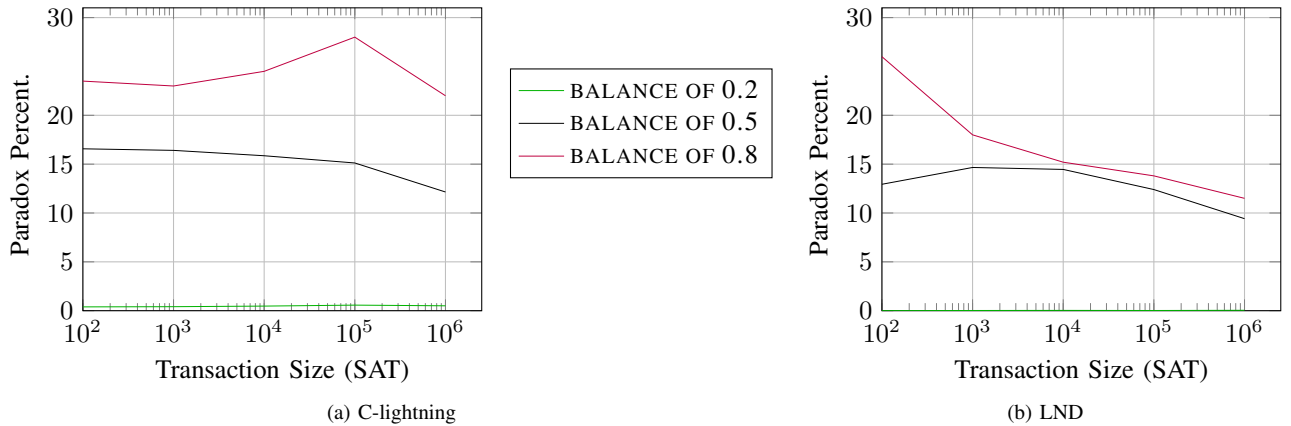


Fig. 11: Percentage of times adding a free-fee link with $TLD = 4$ led to the paradox, for different assumptions on the liquidity distribution on each direction of the link, for: (a) C-lightning, (b) LND.

VI. DISCUSSION: HOW TO MITIGATE THE PARADOX?

Braess Paradox Detection (BPD) problem is algorithmically difficult. Although literature reveals Braess paradox is very

common in real life, the paradox is highly intractable (especially in real networks) and there is no efficient method

to detect the paradox existence [40]. In theory, BPD is an NP-hard problem [41], [42], and designing a network where the paradox does not exist is an NP-hard problem too [43]. The paradox is hard to detect since any changes in network conditions (i.e. add/remove an edge) may lead to a phase of the paradox, or cause it to disappear [40]. Moreover, the paradox relies on users' demand, that is the number of users and the paths they use, and the demand is unpredictable and changes over time. Therefore, studies usually only give heuristics or rules of thumb for BPD, or use some simplifications.

In addition to the difficulties of detecting the paradox in regular networks, BPD in payment channel networks has the difficulty of transactions changing channel liquidity values, thus affecting the entire network, even with no outside intervention. Hence, we provide several rules of thumb for avoiding Braess paradox. We propose rules on dealing with the network topology, though other solutions dealing with payment routing such as using AI-based blockchain [44] or traffic management software (as used to mitigate the paradox in other fields where the paradox appears [45]) for routing might be useful to mitigate the paradox in the future too.

A. Idea 1: Using Ring Topology

We show that the previously analyzed cases of Braess paradox under the liquidity-based fee model and the proportional fee model cannot exist under a Ring topology, as will be shown in Theorem 5. This means that adding a new link to a network such that it becomes structured in a Ring topology cannot cause a fee increase. Similarly, given an existing Ring topology, there does not exist any link that removing it will result in a fee decrease. We start by presenting helpful Lemmas for proving Theorem 5.

Lemma 5. *Under the liquidity-based fee model, with two users transmitting transactions from the same source node to the same destination node. Suppose users employ a strategy leading to Nash Equilibrium state. Denote the paths users choose in Nash Equilibrium state as p_1 and p_2 (p_1 may be the same as p_2). When adding a new link e_{23} , the single-round paradox cannot appear if adding e_{23} does not create a new path that intersects with either p_1 or p_2 .*

Proof. Assume there are two users, u_1 and u_2 , using paths p_1 and p_2 , respectively, in Nash Equilibrium. Towards a contradiction, suppose that adding e_{23} does not create a path that intersects with either p_1 or p_2 , but the paradox does appear. Therefore, adding e_{23} results in a fee increase. Recall that a change in the fee may happen only if one (or more) of the users changed their path, thus we infer it is worth it for one of the users to change her strategy, without loss of generality u_1 , and use a new path p_{new} containing e_{23} . Regarding u_2 's new strategy, we separate the cases:

(i) **Case 1:** In the new equilibrium state u_2 did not change her path. Since p_{new} does not intersect with p_2 , the decision of u_1 to use p_{new} does not affect the fee u_2 pays on p_2 (unless $p_1 = p_2$, in this case u_1 changing her path can only decrease the fee of u_2). In total, the fees of both u_1 and u_2 did not increase, and this is a contradiction to the paradox appearance.

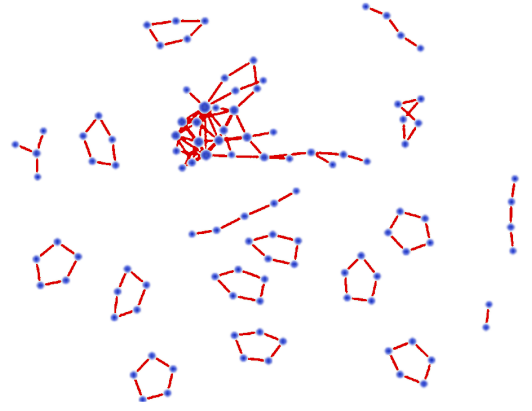


Fig. 12: Raiden network topology graph as of March 2022.

(ii) **Case 2:** In the new equilibrium state u_2 changed her path too. As explained earlier, adding e_{23} does not affect the fee u_2 pays if she does not change her path. Hence, u_2 changes a path only if by using the new path she pays a lower fee. Additionally, when u_2 changes her path the fee u_1 pays cannot be higher than before since otherwise, u_1 could use p_1 and pay the same fee as before, and this would be a contradiction to the equilibrium state the users are in. In total, in this case, the fees of both u_1 and u_2 did not increase, and this is a contradiction to the paradox appearance.

To conclude, upon adding a link that does not create a new path that intersects with either p_1 or p_2 , it is not possible to get a phase of the paradox. $\square$

Lemma 5 was presented with two users for simplicity reasons only and can easily be extended for more users.

Lemma 6. *Consider the proportional fee model with a single user u . In each round, u transmits a transaction from the same source node to the same destination node and follows a greedy strategy. Adding a new link e_{23} that does not create any path intersecting a path that would have been used in the absence of e_{23} cannot cause a multi-round paradox phase.*

Proof. In the multi-round case, the paradox relies on adding e_{23} causing the use of a path with low liquidity links that using it makes other paths unusable and requires u to use a more expensive path. Therefore, when the addition of e_{23} does not create a path that intersects with any of the paths that would have been used had e_{23} not existed, the new path cannot interfere with those previously used paths. As a result, using the new path does not make any of the original paths unusable. Even after the new path is used, u can still access all previously-used paths in each round. Consequently, the fee she pays cannot be higher than before. Thus, the paradox cannot arise in the multi-round case. $\square$

Lemma 6 can be generalized to the case of multiple users too. Intuitively, since every path is independent and is not affected by the usability of other paths, adding a new non-intersecting path cannot cause a fee increase, though we do not prove this lemma here.

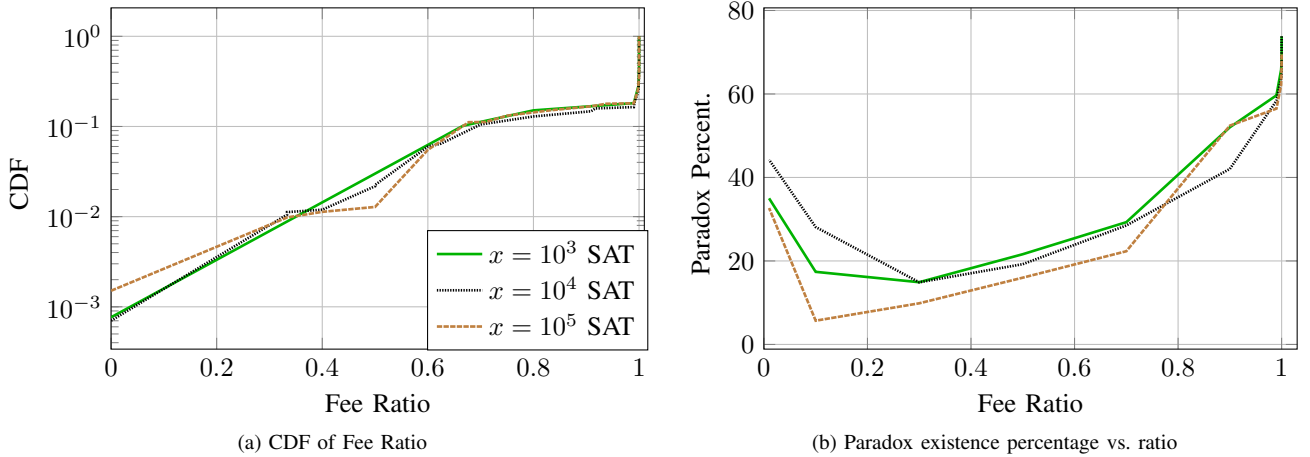


Fig. 13: Paradox statistics on the ratio between the cheapest path fee after and before adding a new link. (a) Ratio CDF in cases where the paradox appears. (b) Paradox existence percentage vs. ratio.

Theorem 5. *In a Ring topology, neither the single-round paradox under the liquidity-based fee model analyzed in Section III nor the paradox under the proportional fee model analyzed in Section IV, can exist.*

Proof. Let e be an arbitrary link on the Ring, and $start$, end arbitrary nodes. We look at the graph without link e and see the impact of adding this link. As without e the graph is a tree, there is only one path from $start$ to end . In particular, adding e cannot create a new path from $start$ to end that intersects with any other path, hence by Lemma 5 and Lemma 6 the paradox cannot appear in either of the cases. $\square$

Using a Ring topology is not only a theoretical proposal, but is also used by real payment channel networks. Raiden [27], for instance, is composed of many graph connected-components that are formed in a Ring topology, as can be seen in Fig. 12. The sketch of Raiden’s topology graph was taken in March 2022, using Raiden Explorer [46]. As we see, Raiden is composed of one main cluster and many other connected components, most of which are in a Ring topology. Theorem 5 proves that in every such connected component formed in a Ring topology, there is no link whose appearance causes a fee increase.

B. Idea 2: Restrictions on New Paths

Another interesting rule of thumb concerns restricting the addition of new channels to the network under certain conditions. In some cases, adding a new cheap link does not decrease the fee much, and makes other paths become unusable faster. We performed an analysis to support this idea and present a possible restriction rule that can be used. The analysis is very similar to the Lightning paradox analysis presented in Section V: We chose arbitrary source and destination nodes, calculated the cheapest paths between these nodes (with a path length limitation of three channels), and looked at the consequences of adding a new free-fee link that connects the two cheapest paths.

When adding a new link, we look at the ratio between the old path fee (the fee of the cheapest path before adding a new link) and the new path fee (the fee of the cheapest path after adding the link). Fig. 13 presents insights into the ratio between the new path fee to the old path fee. Only cases where the ratio is smaller than one were examined, meaning that adding a new link created a cheaper path than any other existing path. The analysis examined three transaction sizes: $x = 10^3, 10^4$ and 10^5 SAT. First, we wanted to analyze the distribution of the ratios when the paradox appeared as a result of the link addition, and find what ratios make up the majority of paradox appearances. Fig. 13a presents the CDF of links whose adding causes the paradox appearance compared to the fee ratio. More than 85% of the paradox appearances appear when the ratio is 0.84 or higher. Moreover, 74-78% of the paradox appearances happen when the ratio is higher than 0.999, i.e. when the new path created by adding the new link is cheaper by only 0.1% or less. Fee ratios of 0.9999999 or higher make up 58-64% of the paradox appearance cases.

After analyzing the ratio distribution given the paradox already exists, we analyzed the chances of the paradox appearing for different ratio sizes. Fig. 13b presents the paradox existence percentage compared to the fee ratio. Interestingly, when adding a link having a ratio of 0.9999999 or higher, the chances of the paradox appearing were 69-74%. Surprisingly, small ratio values (less than 0.1) had higher chances of the paradox appearing ($\approx 35\%$) compared to higher ratio values (0.3-0.7, $\approx 25\%$). We attribute this to the fact that although the new paths created were much cheaper (since when creating low-liquidity channels less money is locked, we assume lower fees are charged when using low-liquidity channels), they had much less liquidity compared to paths created in cases with higher ratio values. Hence, expensive paths were needed to be used relatively quickly. Yet, the paradox had the highest chance of appearing for high ratio values, and these percentages increase the closer the ratio is to 1.

From this analysis, we infer that in some cases a payment channel network should disallow the opening of new payment

channels. As shown earlier, payment channel networks tend to serve recurring transactions, some of them a significant number of times. By focusing on high-traffic nodes and prohibiting the creation of new channels in some cases, the paradox may be mitigated. For instance, given important source and destination nodes (e.g., if many transactions are transmitted between these nodes), if Lightning prohibited adding new links that connect the current two cheapest paths and create a new path with a fee ratio of 0.9999999 or higher, it would be able to decrease its paradox appearances by more than 58% for these nodes. Since in about 70% of cases adding such links causes a paradox appearance, this restriction would mainly prevent the paradox appearance, and would not limit the addition of links that in practice decrease the fee.

C. Future Directions

Beyond characterizing when and how the Braess paradox arises in PCNs, our results suggest several directions for dampening its impact. We highlight three complementary approaches: modifying routing behavior, designing topologies that are inherently more robust, and changing how payments are split across paths.

a) Path Randomization: A promising mitigation direction is the use of randomized or load-balancing routing strategies that deliberately spread traffic across multiple near-optimal paths, rather than always selecting a single greedy shortest path. This type of strategy is used in Flash [13]. By reducing the concentration of flow on any one low-liquidity channel, such schemes may slow or dampen the liquidity-depletion dynamics that underlie the multi-round Braess paradox, even if they cannot fully eliminate it. Systematic analysis of how different randomized policies interact with fee structures and liquidity evolution remains an open problem for future work.

b) Clustering: Section VI-A constructs PCNs in the form of multiple ring-based graphs to avoid the paradox. Beyond ring topologies, various other graph constructions can be used to mitigate Braess-like effects. Our recent work [47] explores hyperedge-based graph topologies that address liquidity issues in PCNs, increasing transaction success rates and allowing execution with lower channel liquidity. An interesting direction for future work is to study whether such clustered and hyperedge-based constructions can provably limit or eliminate the multi-round Braess paradox in our setting.

c) Micropayments: Another potential mitigation avenue is the use of micropayment-style schemes, such as Spider [14], that split each payment into many small units and distribute them over multiple paths. By more evenly balancing flow and liquidity across the network, such approaches may reduce the risk that a new “shortcut” channel is overused and prematurely depletes its liquidity, thereby softening the multi-round Braess effect. Quantitatively characterizing how flow-splitting interacts with different fee models and liquidity dynamics in our setting is an interesting direction for future work.

VII. CONCLUSIONS AND FUTURE WORK

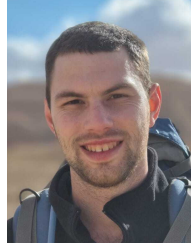
This research studies the potential existence of Braess paradox in PCNs and analyzes the conditions for its appearance

under two different fee models: Liquidity-based fees and proportional fees. We provided indications for the appearance of the multi-round paradox in the Lightning network based on its real topology. Last, we discussed optional ways to mitigate the paradox, either with the use of the ring topology or through restrictions on new paths. Future work includes evaluating additional mitigation mechanisms—such as randomized routing, clustered topologies, and micropayment-based flow-splitting—both analytically and on real-world PCN data. Additionally, we intend to investigate how malicious behavior can be used to increase network fees.

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